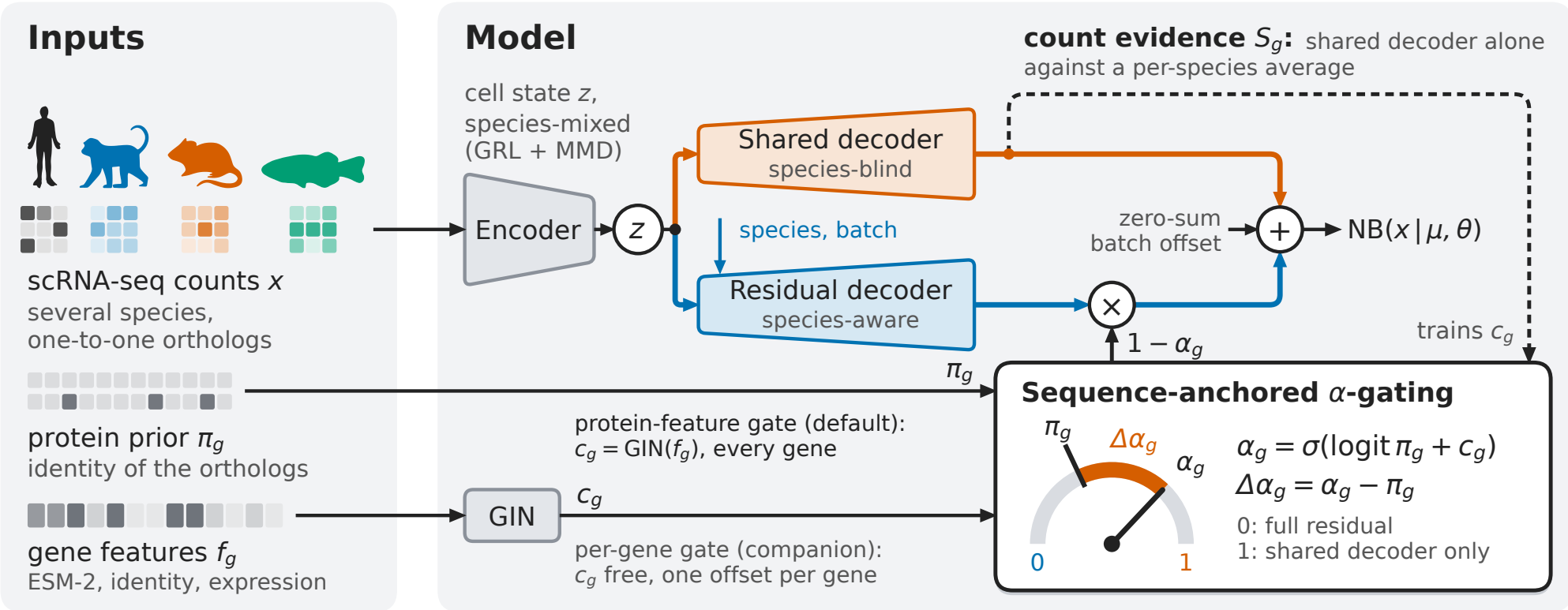
Inputs
scRNA-seq counts x
several species,
one-to-one orthologs
protein prior π_g
identity of the orthologs
gene features f_g
ESM-2, identity, expression
Model
cell state z,
species-mixed
(GRL + MMD)
Encoder
z
Shared decoder
species-blind
species, batch
Residual decoder
species-aware
count evidence S_g: shared decoder alone
against a per-species average
zero-sum
batch offset
+
NB($x \mid \mu, \theta$)
×
$1 - \alpha_g$
trains c_g
π_g
Sequence-anchored α-gating
π_g
$\Delta\alpha_g$
α_g
0
1
$\alpha_g = \sigma(\text{logit}\,\pi_g + c_g)$
$\Delta\alpha_g = \alpha_g - \pi_g$
0: full residual
1: shared decoder only
protein-feature gate (default):
$c_g = \text{GIN}(f_g)$, every gene
GIN
c_g
per-gene gate (companion):
c_g free, one offset per gene